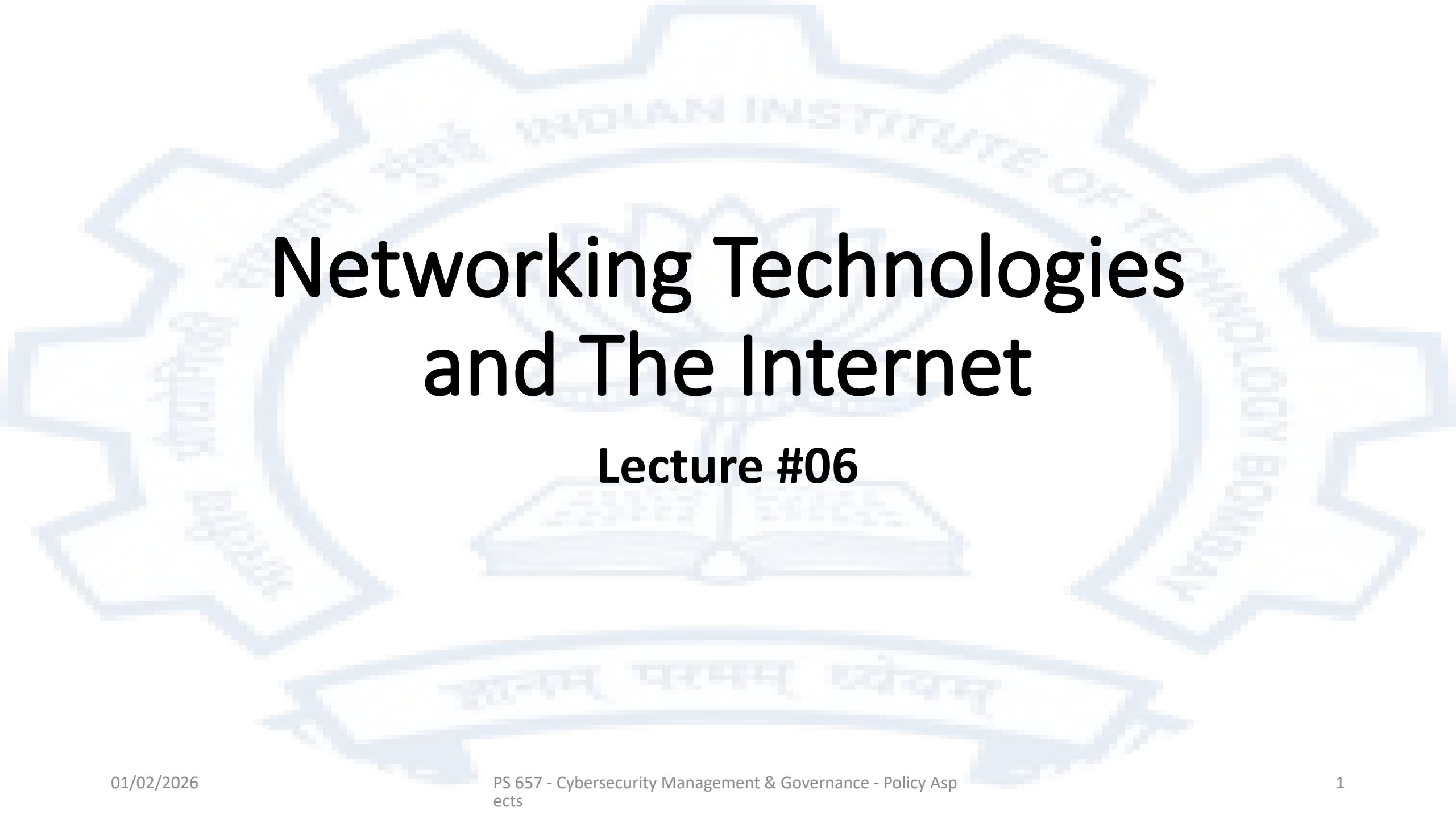
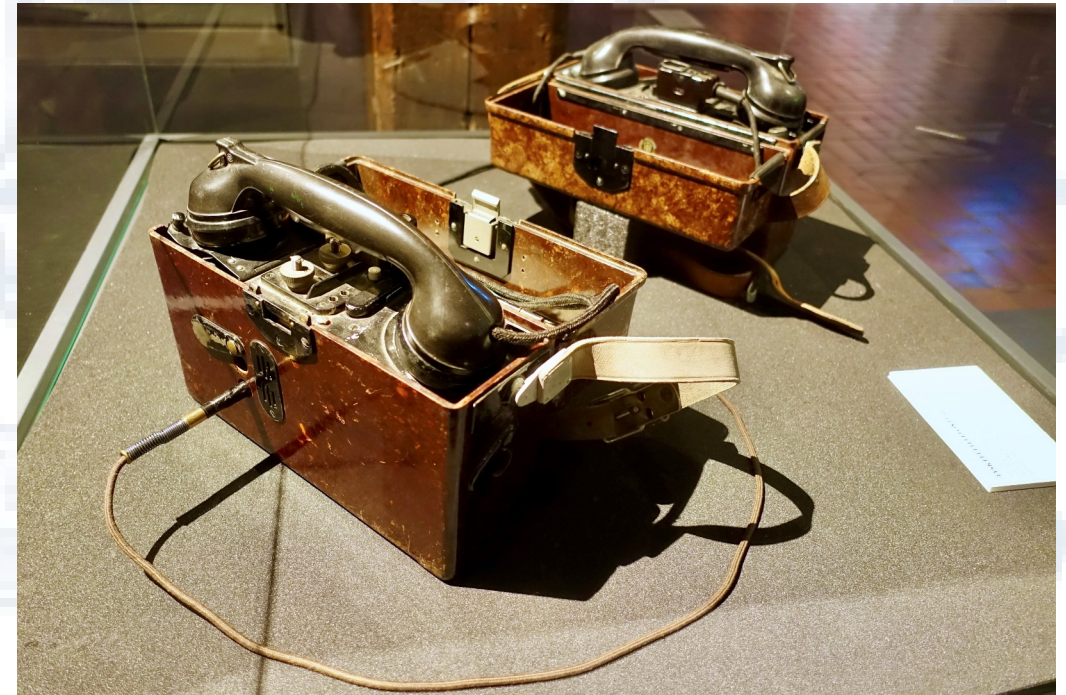
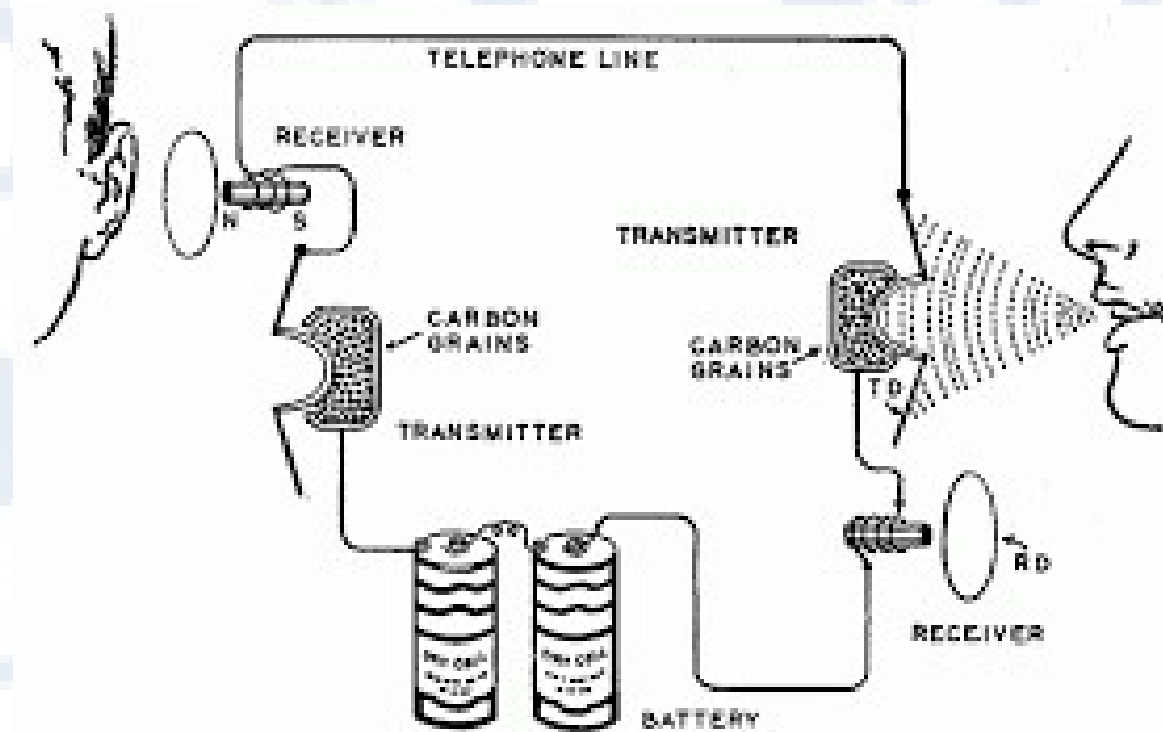




Networking Technologies and The Internet

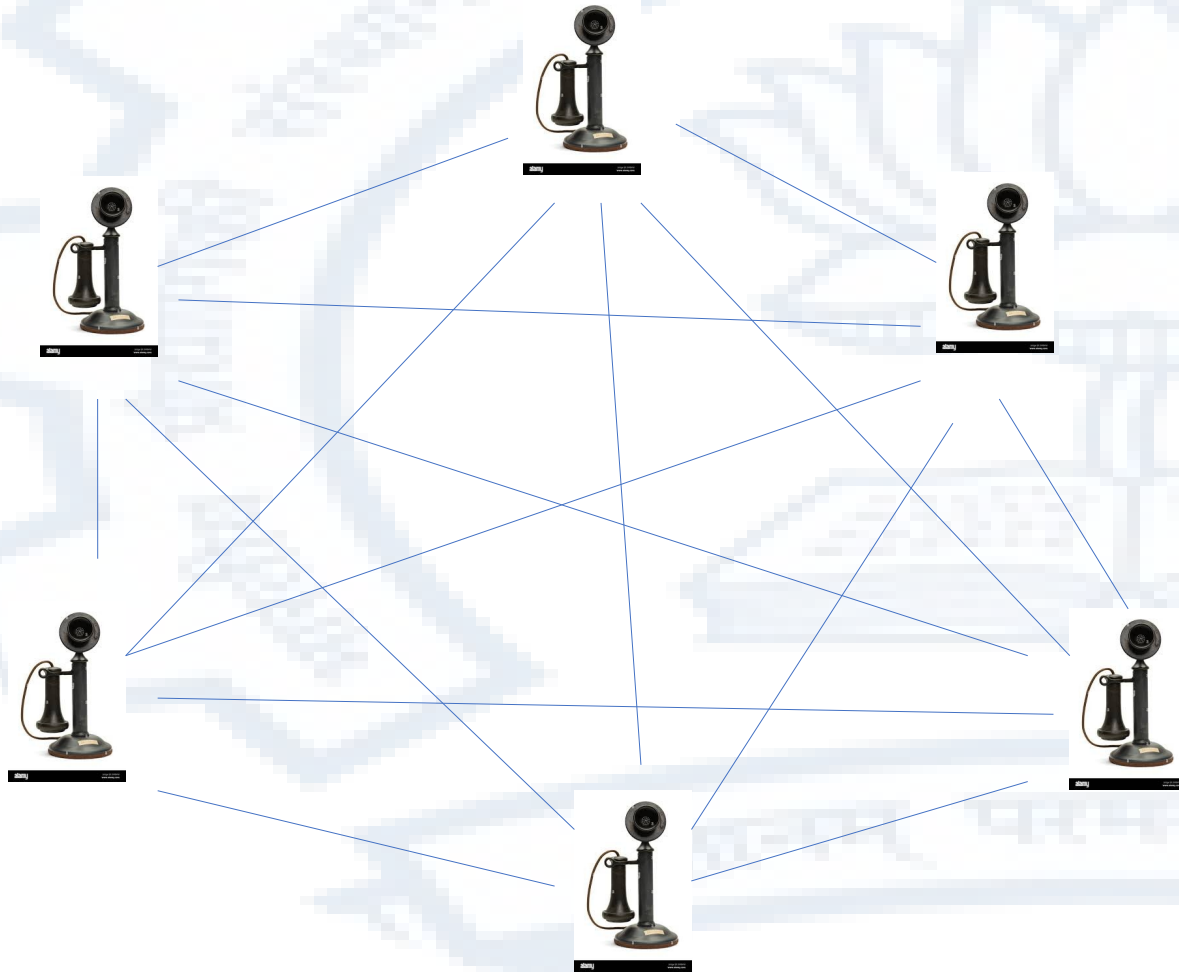
Lecture #06

Telecommunication



The earliest phone was a point-to-point system

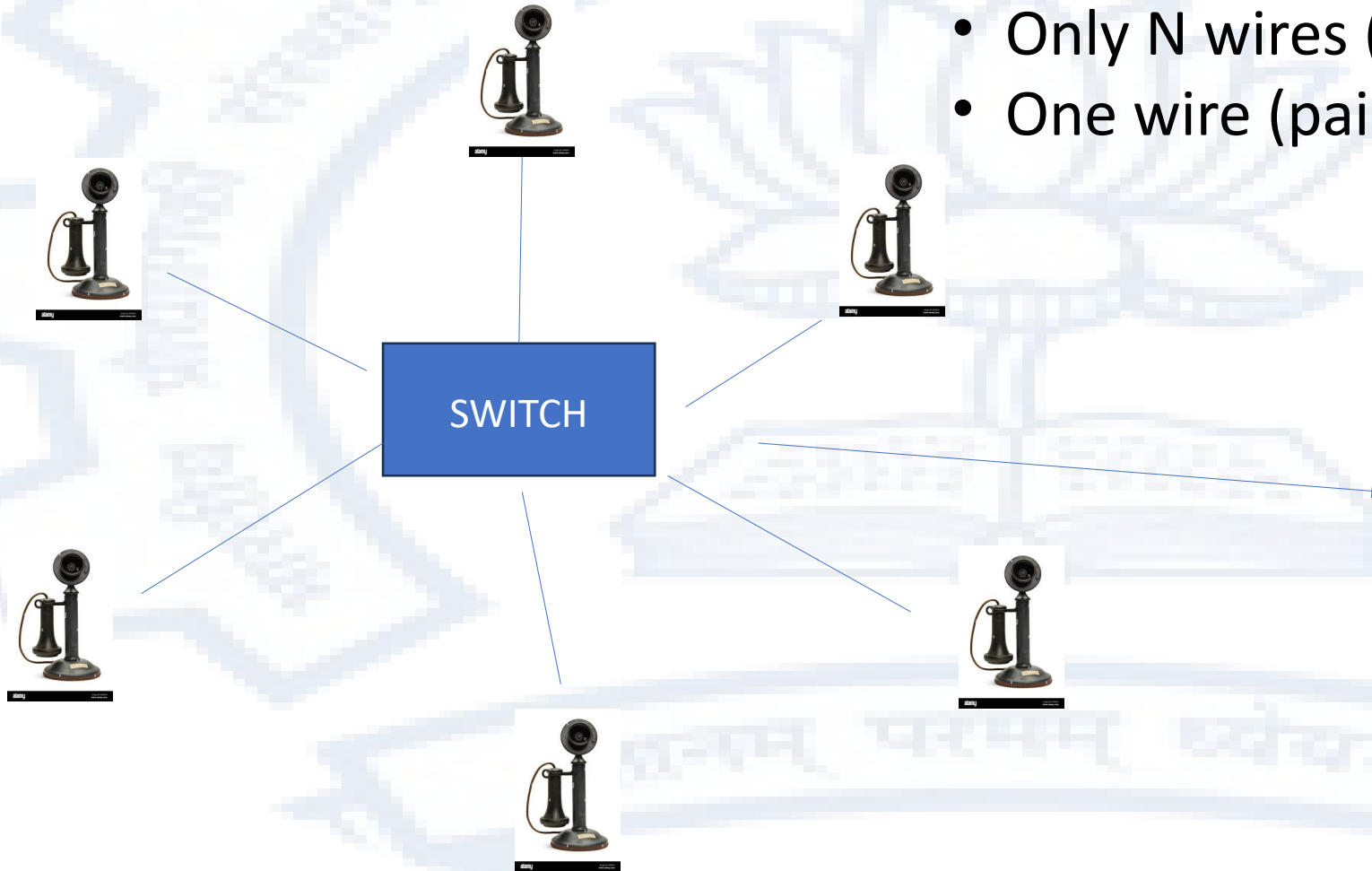
How do you talk to many people?



- Too many wires
- N people means $\{N \times (N-1)\} \div 2$
- If there are 1000 people we need 4,99,500 (pairs) of wires !!
- This is impractical

A better method

- Only N wires (pairs)
- One wire (pair) per participant (phone)



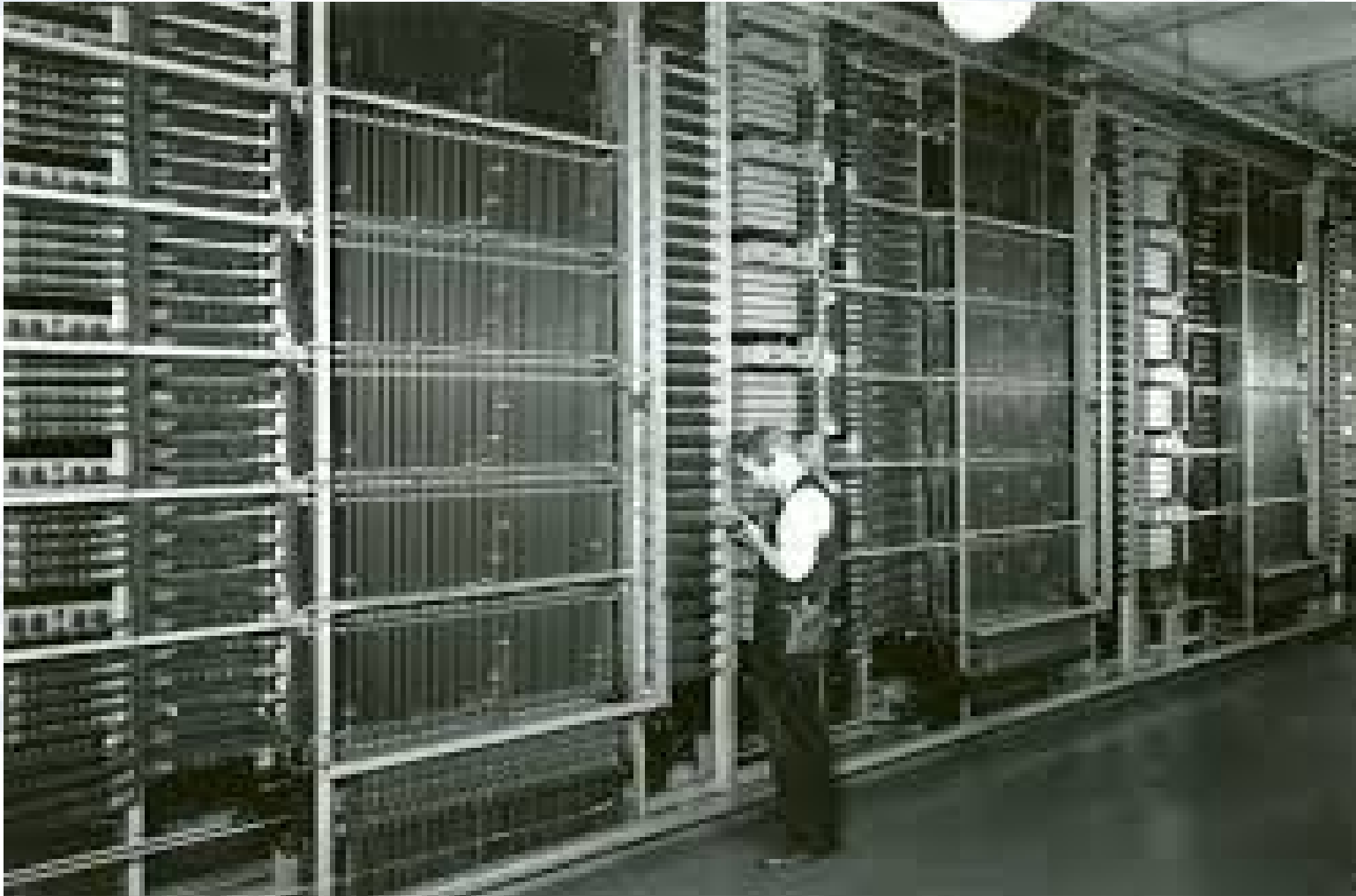
What is this SWITCH?

The Earliest SWITCH- A switchboard

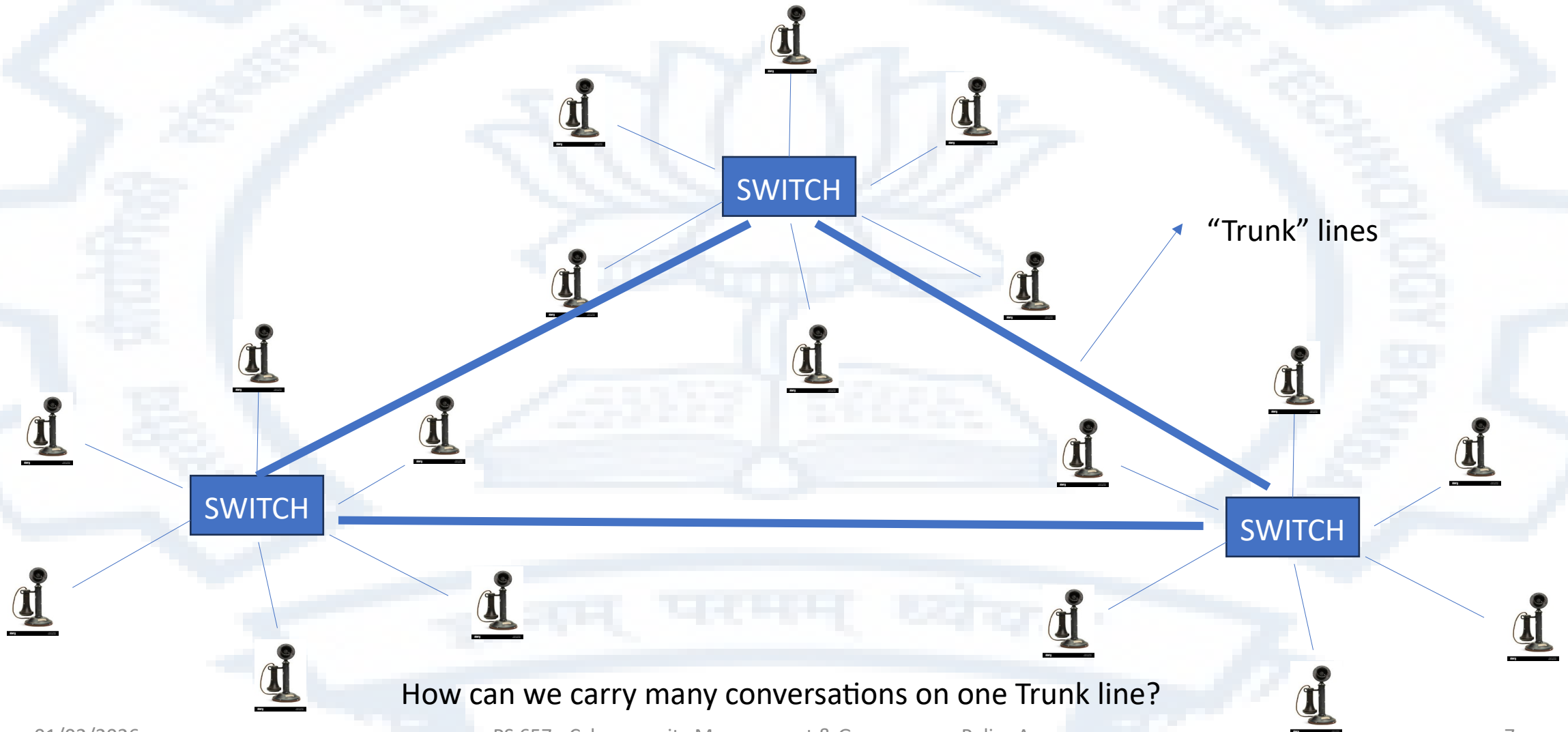


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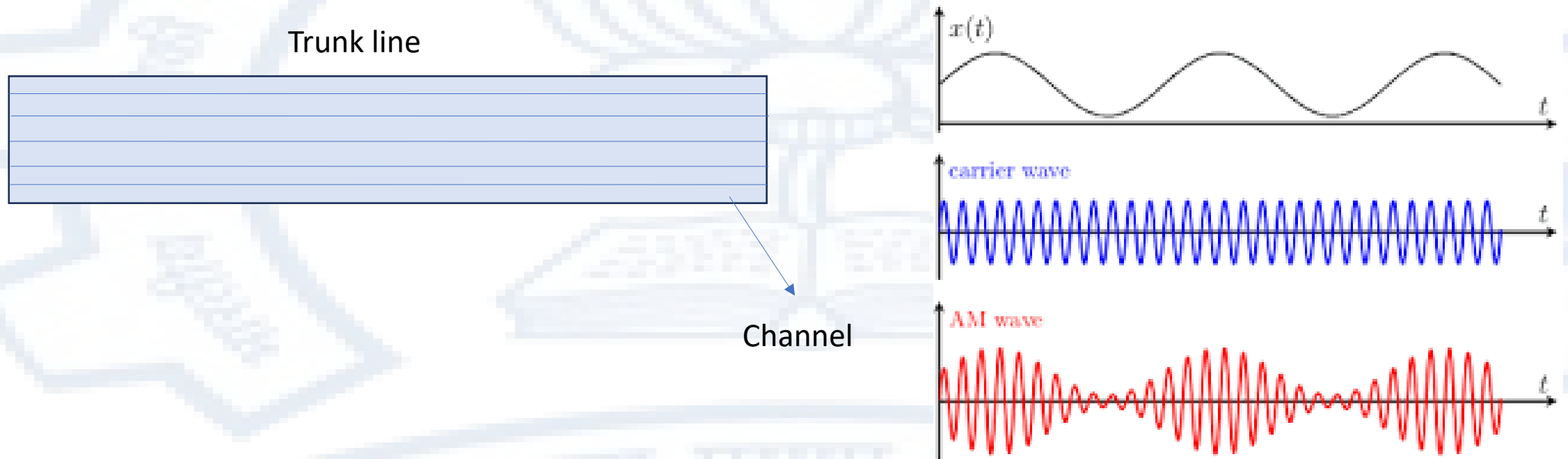
A Classical Automated Switch



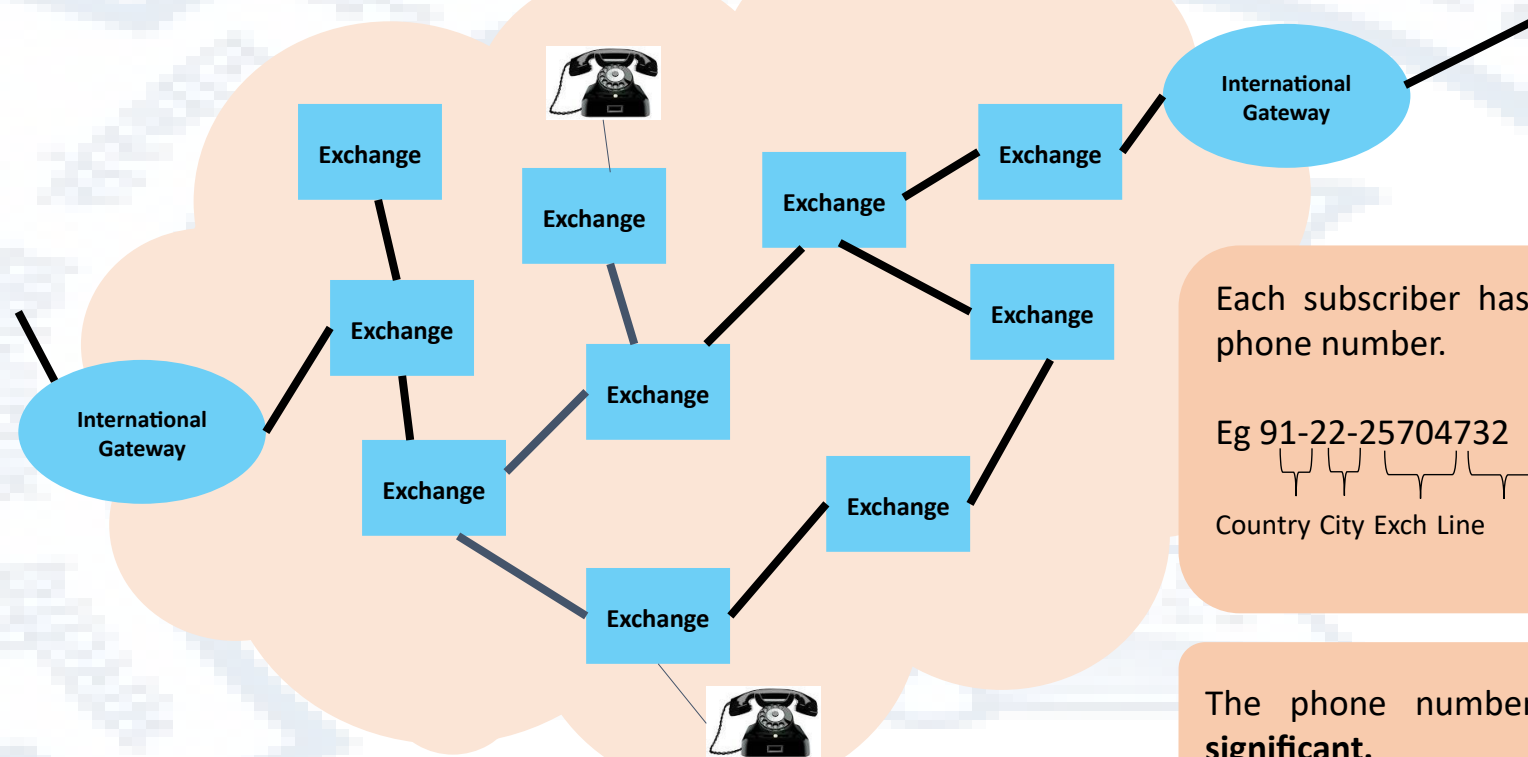
Interconnecting Switches



Carrying multiple conversations on one physical wire



Traditional Telecommunication Systems



Each subscriber has a **globally unique** phone number.

Eg 91-22-25704732

Country City Exch Line

The phone number is **geographically significant**.

- Calls originating from one country to another must pass via an International Gateway.
- The international numbering conventions, country codes, standards etc. are decided by **ITU which is a specialized agency of the UN.**

The system creates a **circuit** (opto-electronic path) from caller to recipient which is **maintained for the entire duration of the call.**

The Problems with using conventional telecommunication systems for computer to computer communication

- **Inefficiency with Data Transmission:** Circuit-switched networks establish a dedicated communication path between two points for the duration of the session. This is inefficient for computer communication, as data transmission is often bursty and doesn't require a constant, dedicated connection, but when a connection is needed it is a high bandwidth need
- **Resource Wastage:** Because the circuit remains open regardless of whether data is being transmitted or not, it leads to underutilization of network resources. This is in contrast to data communication, where resources should be dynamically allocated.
- **Scalability Issues:** Circuit-switched systems are not easily scalable to handle the large number of connections typical in computer networks. Each connection requires a separate dedicated circuit, which is impractical for networks with many simultaneous data sessions.
- **Higher Latency:** Establishing a circuit-switched connection can introduce latency, which is undesirable for many real-time computer communication applications.
- **Cost:** Maintaining a circuit-switched network for data communications can be more expensive due to the need for constant, dedicated pathways.
- **Limited Flexibility:** Circuit-switched networks are less adaptable to varying data loads and can't easily prioritize different types of data, such as video versus simple text.

Packet Switching

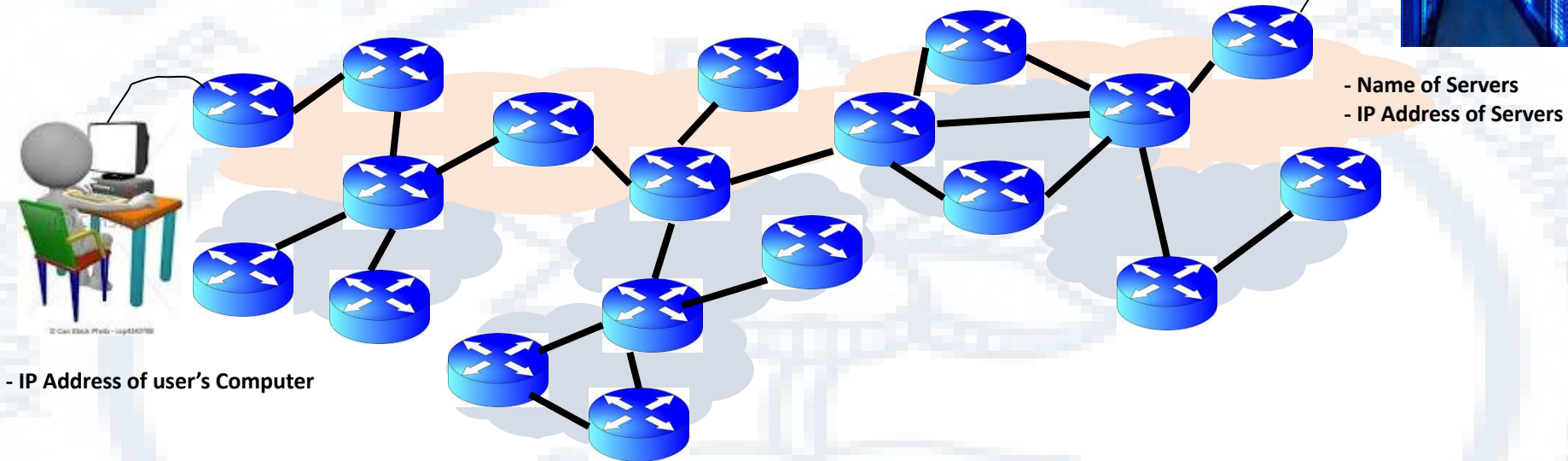
- Leonard Kleinrock of MIT first described the concept of packet switching in a paper in 1961
- The concept of packet switching formed a very important part of Roberts' vision for ARPANET which was the precursor to the Internet
- The basic concept was that streams of data were broken into smaller fixed sized chunks called “packets” and the network switched “packets” rather than circuits

Internet – The Early Days

- In 1985 the US National Science Foundation (NSF) began to provide various universities connectivity to certain supercomputer sites. This network used all the new technologies including TCP/IP that had been proved in ARPANET.
- By 1991 NSFNET had deployed a good backbone network and provided connectivity to many universities and research institutions
- The Internet, as it had started to be called, was a non-commercial entity for promoting research and education.

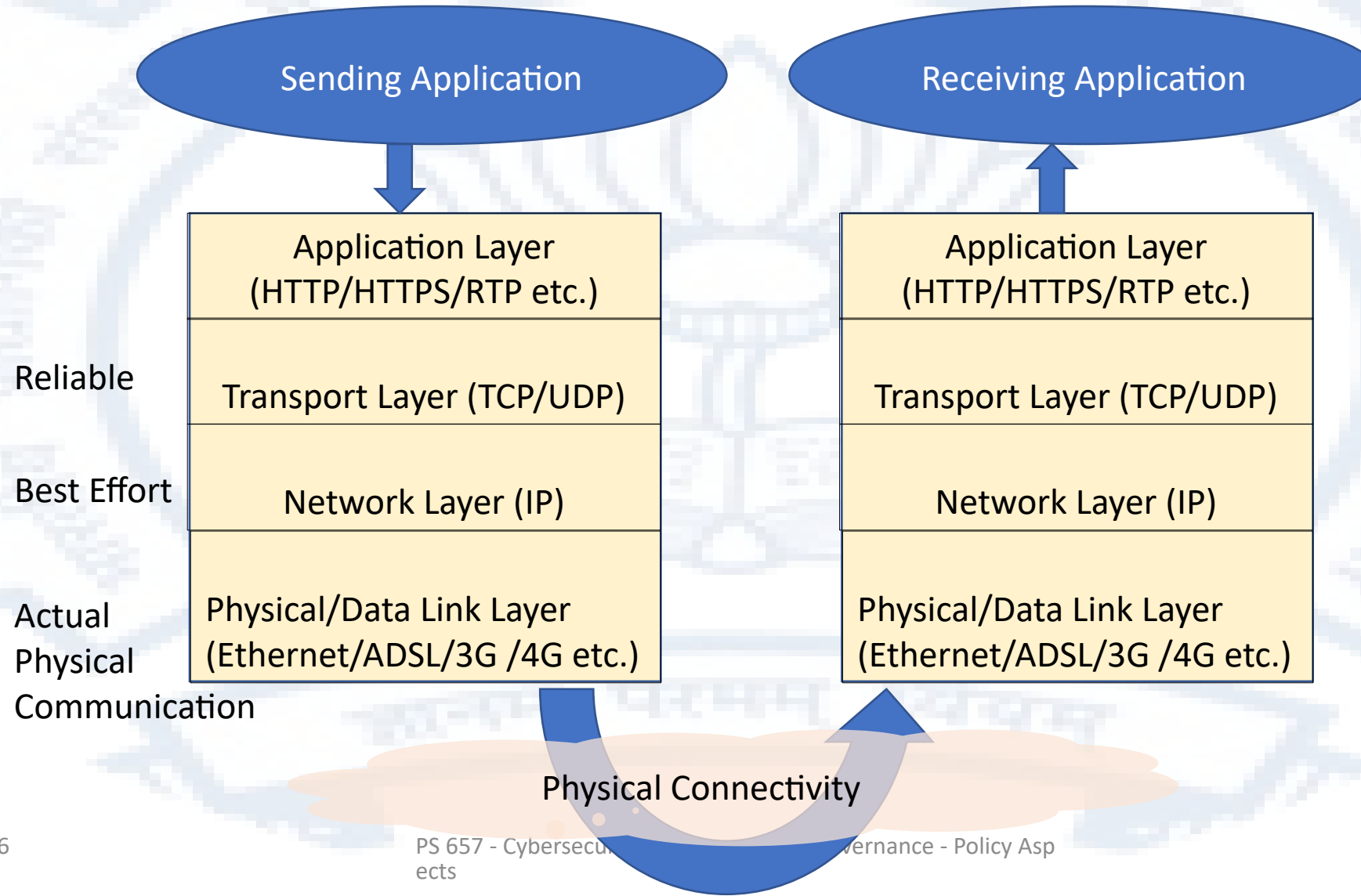
The Internet

An International Network of Networks (Packet Switched)

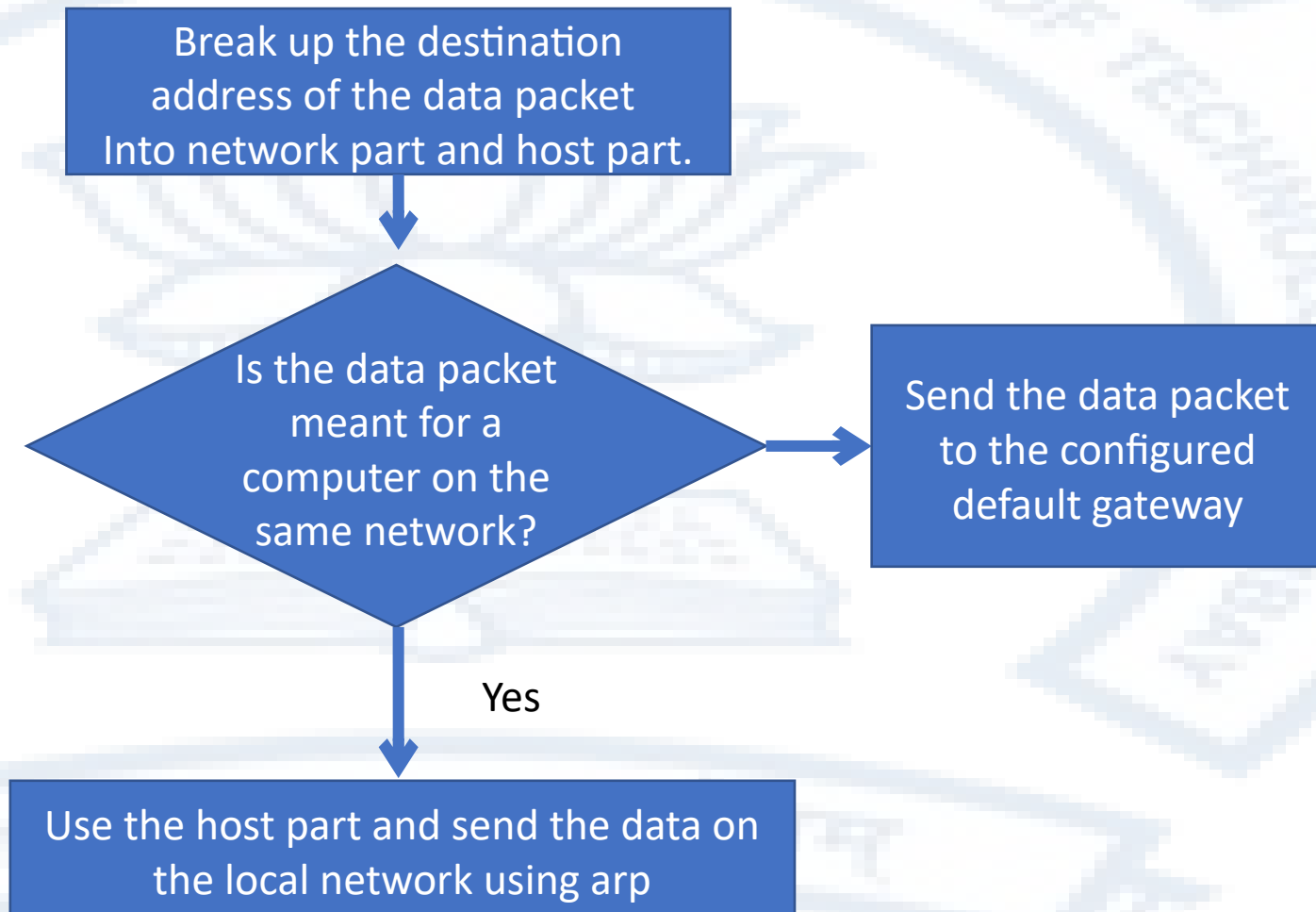


- Each computer or mobile device is identified by its IP address.
- IP addresses need to be globally unique. They consist of 4 octets, eg **121.32.7.25** etc.
- The basic technology provides a method for sending a “packet” of data from one IP address to another IP address.
- It is not convenient to remember the IP address of every server. Therefore servers are identified by names which are easier to remember such as www.google.com, www.wikipedia.org etc.
- The two basic functions of the Internet are conversion of names to IP addresses (DNS) and getting a packet from one IP address to another IP address (Routing).
- The allocation of IP address ranges to various Internet Service Providers and the running of the root DNS service is done by ICANN which is a not-for-profit private sector company incorporated in Los Angeles USA.
- **IP addresses do not have a very strong geographic significance.**

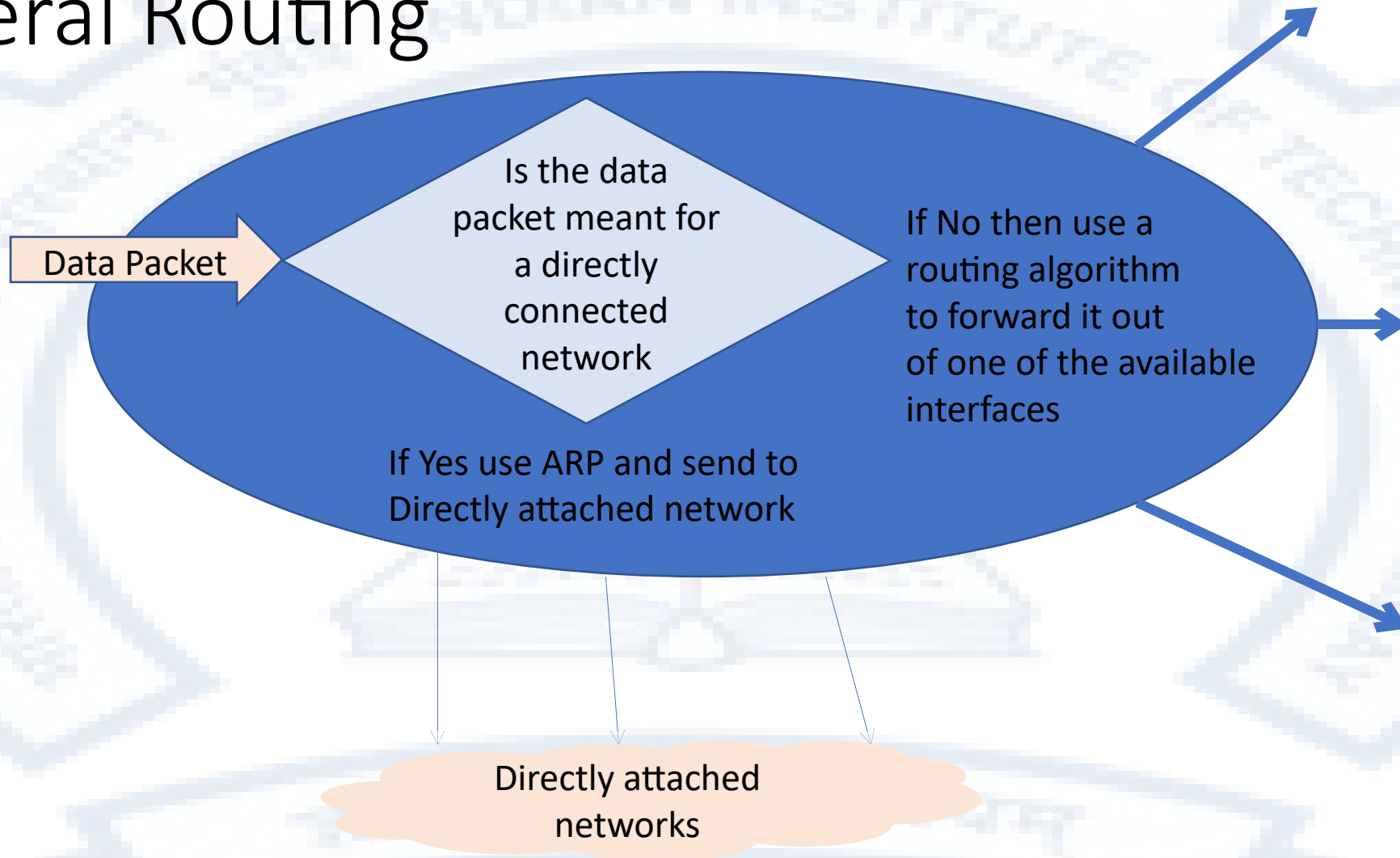
Concept of layered communication



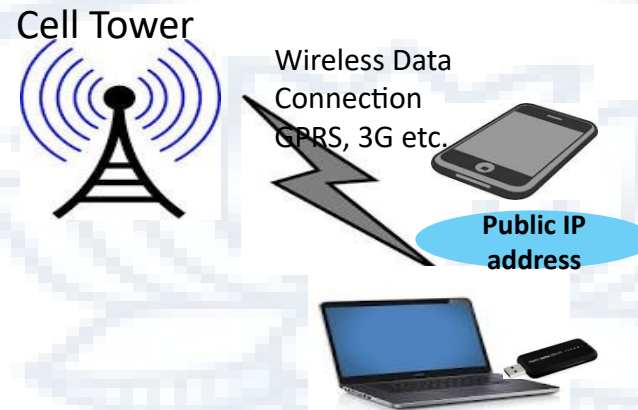
Routing from a local PC



General Routing

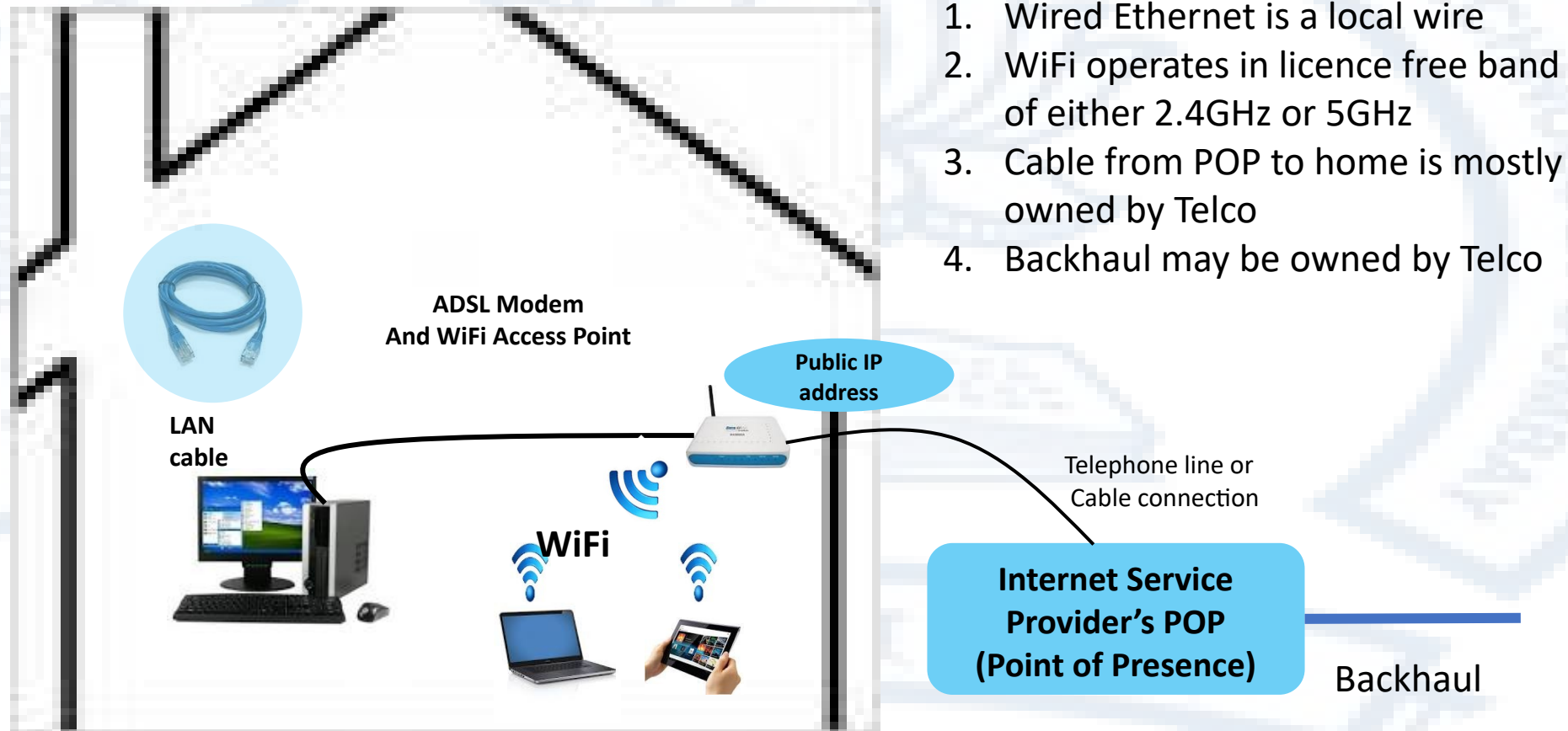


Individual Mobile User Access

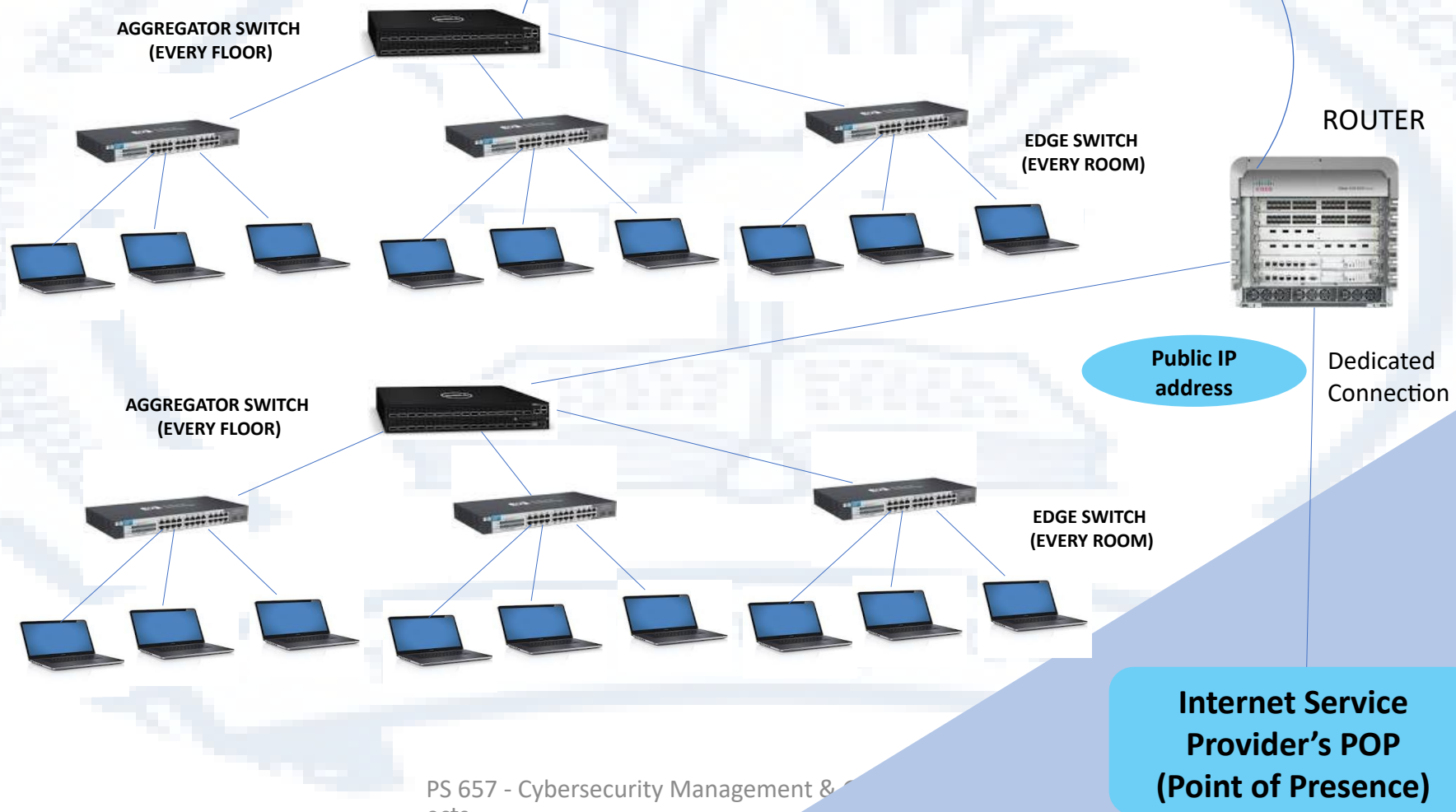


1. Each mobile device has a unique IMEI /MEID number which is set by the manufacturer
2. Each mobile phone needs a SIM card which has a unique number and a programmable phone number. Before they are allowed to give a user a SIM card they must collect one identity proof and one address proof. The mobile provider thus has a list of all subscribers names ,SIM (IMSI) numbers and mobile numbers and the IMEI number that a currently being used SIM is associated with.
3. The mobile service providers have paid license fees to the Government, and therefore it can be considered that the mobile service provider “owns” the last leg of connectivity to the user. The 800MHz, 900MHz and 1800MHz bands are in use for 2G, 3G and 4G

Connecting to the Internet – Home Users



Connecting to the Internet – Large Organizations



Satellite Links

- These are convenient to use in remote areas
- However they do not provide the level of bandwidth that terrestrial and submarine cables can provide
- Coordination done by ITU which is a specialized UN agency

Terrestrial Cable Networks

Wholesale Bandwidth Providers in India

- Tata Communications
- Reliance
- Airtel
- BSNL – NIB
- Sify
- Dishnet
- Pacnet

Communications Companies hire bandwidth from bandwidth providers for

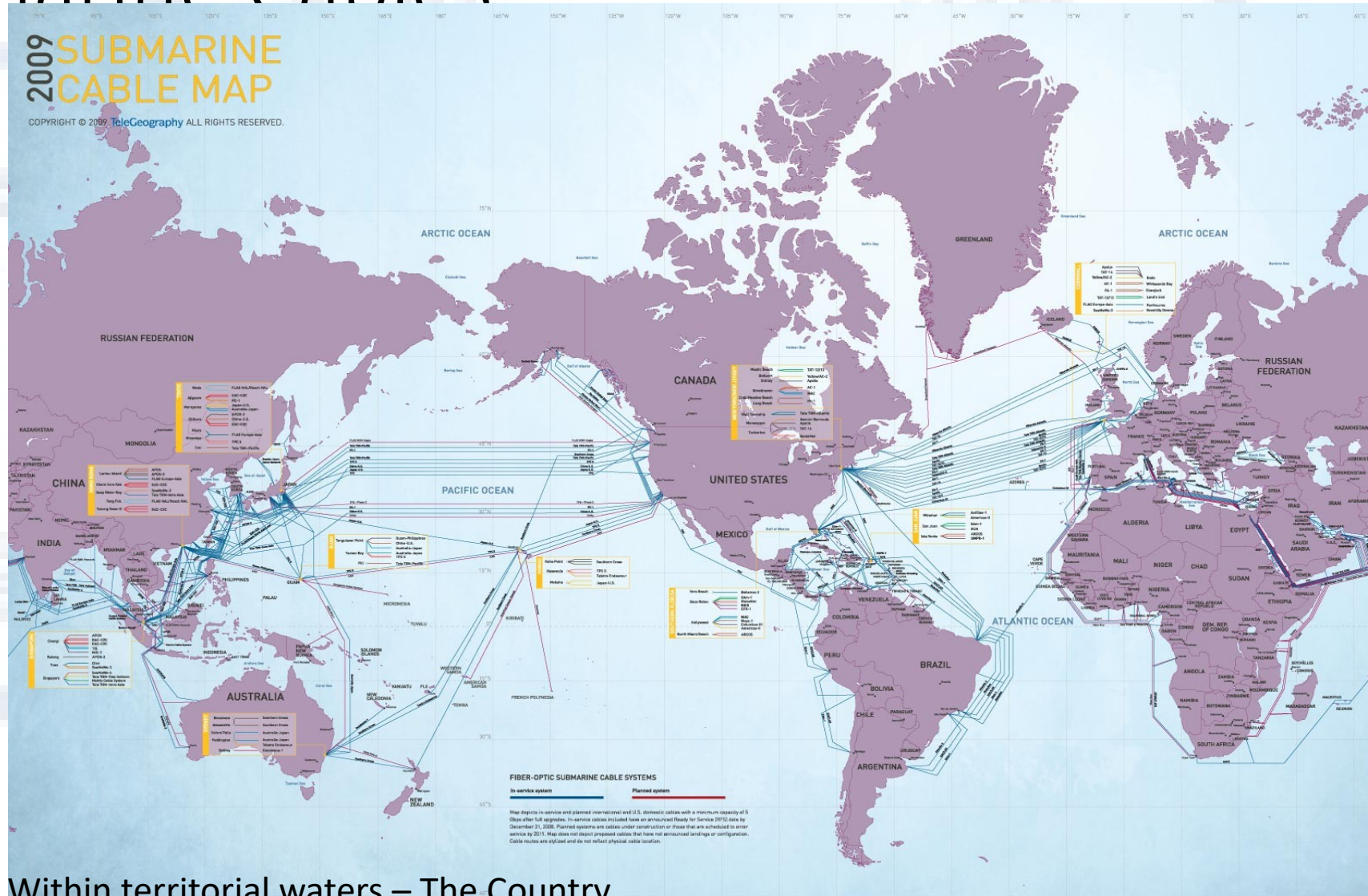
- **Connecting their own Infrastructure (backhaul)**
- **Last mile connectivity to customers**

Many companies provide
Retail bandwidth in various
Geographical areas

Coordination is the responsibility of National Government Agencies

- Telecom licence
- Right of way
- Cable laying and maintenance

Submarine Cables



- Within territorial waters – The Country
- Otherwise UNCLOS

IP Address

- Every computer, mobile phone or device that connects to the Internet has a “IP Address”
- This IP Address is similar to a phone number in the conventional telephone system
- There are 2 types of IP address
 - IPv4 : These are 32 bits and written in dotted notation such as 172.16.254.1 (2^{32} distinct addresses – 4,29,967,296 – 42 crore)
 - IPv6 : These are 128 bits and an example of an IPv6 address is 2001:db8:0:1234:0:567:8:1 (2^{128} distinct addresses – very very many)

Structure of an IPv4 Address

- These are 32 bits and written in dotted notation such as 172.16.254.1
- There are two parts to any IP address these are
 - Network part
 - Host part (identifies the specific computer)

Class A IPv4 Address

- These are addresses of the form
1.x.y.z to 126.x.y.z (first bit of 1st octet is 0 – first here is MSb)
- The network part is the first octet followed by zeroes. Thus Class A networks are 1.0.0.0 to 126.0.0.0
- The host part is the balance three octets, i.e. x.y.z
- The default netmask for Class A addresses is 255.0.0.0

Class B IPv4 Address

- These are addresses of the form
128.0.x.y to 191.255.x.y (first two bits of first octet are 10)
- The network part is the first two octets followed by zeroes. Thus Class B networks are 128.0.0.0 to 191.255.0.0
- The host part is the balance two octets, i.e. x.y
- The default netmask for Class B addresses is 255.255.0.0

Class C IPv4 Address

- These are addresses of the form
192.0.0.x to 223.255.255.x (first three bits of first octet are 110)
- The network part is the first three octets followed by zeroes. Thus Class B networks are 192.0.0.0 to 233.255.255.0
- The host part is the balance one octet, i.e. x
- The default netmask for Class B addresses is 255.255.255.0

Special Classes of IPv4 Addresses

- Class D – 224.0.0.0 to 239.255.255.255
 - These do not have any host part and are used for multicasting only
- Class E – 240.0.0.0 to 255.255.255.254
 - There are for experimental and R&D purposes only and also do not have any host part
- 255.255.255.255 – The broadcast address
- 127.0.0.0 – For the local-host network
- 10.0.0.0 – A private Class A network
- 172.16.0.0 to 172.31.0.0 – 16 contiguous private class B networks
- 192.168.0.0 to 192.168.255.0 – 256 contiguous private class C networks

Structure of an IPv6 Address

- IPv6 addresses are longer and can address very many more endpoints
- IPv6 addresses are conceptually simpler than IPv4 addresses

Example

2001:0db8:0000:1234:0000:0567:0008:0001

First 6 bytes = 48 bits
are the network address

Next 2 bytes = 16 bits
are the subnetwork
address

Last 8 bytes = 64 bits
are the host/device/endpoint
address

Each group is 2 bytes = 4 octets
= 16 bits
The whole address is 16 bytes =
32 octets = 128 bits

Routing

- The basic purpose of the routing function is to make sure that a packet can be sent from one IP address (source) to another IP address (destination). This is done using the IP protocol
- The basic IP routing function is done on a “best effort” basis. Packets may be lost
- In order to achieve guaranteed delivery of data packets, a higher layer protocol called TCP is used
- Together the terminology TCP/IP is used to describe how data is routed on the Internet

Configuring IP addresses

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\170339>ipconfig

Windows IP Configuration

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wireless Network Connection 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wireless Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wireless Network Connection:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::f0f9:e7e8:3ad3:1249%11
    IPv4 Address. . . . . : 10.17.88.139
    Subnet Mask . . . . . : 255.255.252.0
    Default Gateway . . . . . : 10.17.88.1

Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Tunnel adapter Local Area Connection* 11:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Tunnel adapter isatap.{46CAD07F-11FC-474B-BE44-89893745A7C1}:
```

Structure of a Domain Name

www.incometaxindia.gov.in → ccTLD

www.google.com

en.wikipedia.org

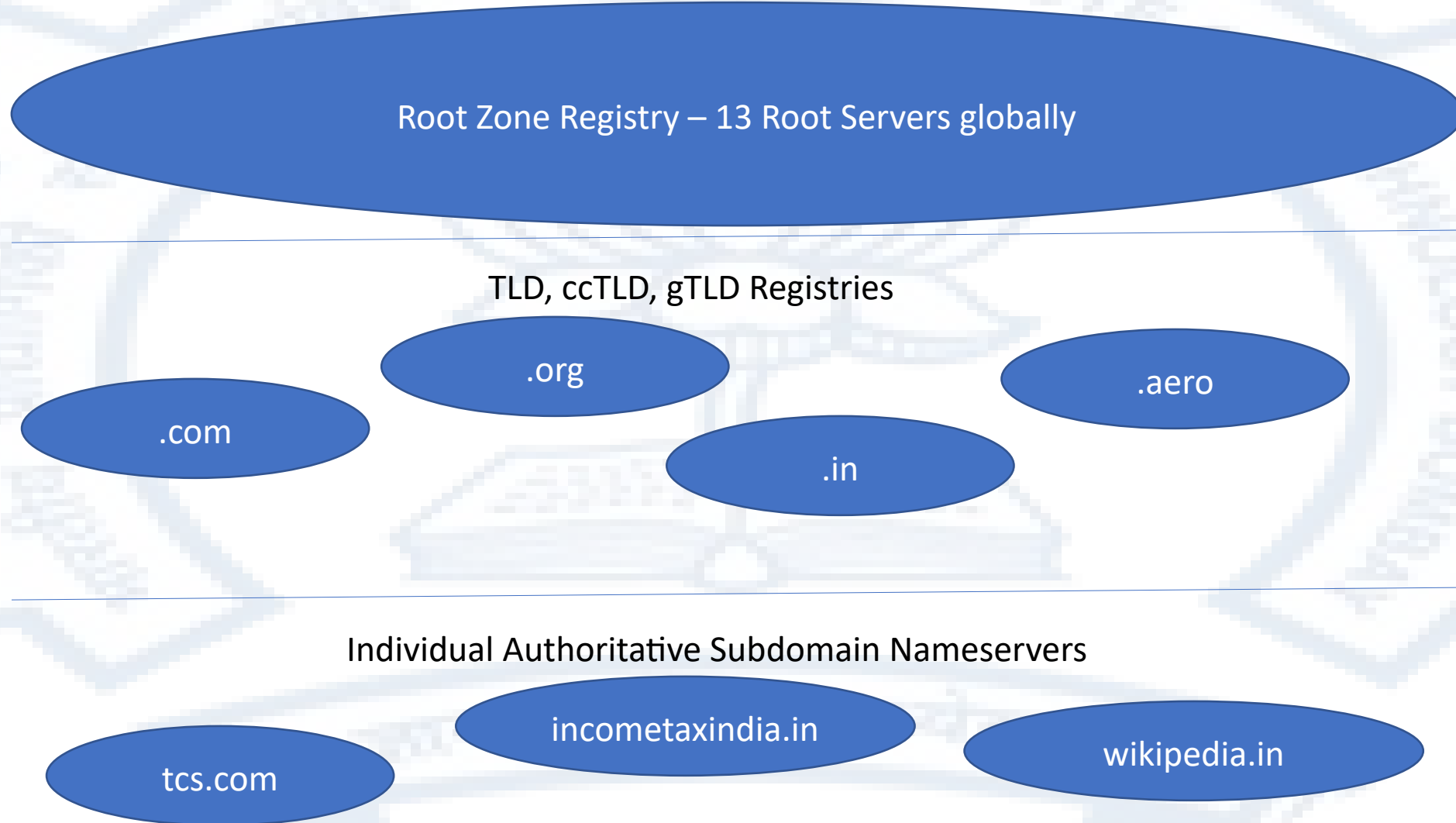
www.ultimatix.net

mta.info

gTLD

TLD

Hierarchical Nature of DNS



Example of DNS resolution

Obtaining Root Servers from localhost

```
$ dig +trace mozg.mit.edu
.      516232 IN  NS  i.root-servers.net.
.      516232 IN  NS  g.root-servers.net.
.      516232 IN  NS  b.root-servers.net.
.      516232 IN  NS  a.root-servers.net.
.      516232 IN  NS  j.root-servers.net.
.      516232 IN  NS  f.root-servers.net.
.      516232 IN  NS  k.root-servers.net.
.      516232 IN  NS  m.root-servers.net.
.      516232 IN  NS  e.root-servers.net.
.      516232 IN  NS  h.root-servers.net.
.      516232 IN  NS  d.root-servers.net.
.      516232 IN  NS  l.root-servers.net.
.      516232 IN  NS  c.root-servers.net.
;; Received 508 bytes from 127.0.0.1#53(127.0.0.1) in
863 ms
```

Example of DNS resolution

Obtaining .edu servers from 199.7.83.42

edu.	172800	IN	NS	a.edu-servers.net.
edu.	172800	IN	NS	c.edu-servers.net.
edu.	172800	IN	NS	d.edu-servers.net.
edu.	172800	IN	NS	f.edu-servers.net.
edu.	172800	IN	NS	g.edu-servers.net.
edu.	172800	IN	NS	l.edu-servers.net.

;; Received 265 bytes from 199.7.83.42#53(199.7.83.42) in
1847 ms

Example of DNS resolution

Obtaining mit.edu servers from 192.35.51.30

```
mit.edu.      172800  IN   NS   usw2.akam.net.  
mit.edu.      172800  IN   NS   asia1.akam.net.  
mit.edu.      172800  IN   NS   asia2.akam.net.  
mit.edu.      172800  IN   NS   use2.akam.net.  
mit.edu.      172800  IN   NS   ns1-37.akam.net.  
mit.edu.      172800  IN   NS   ns1-173.akam.net.  
mit.edu.      172800  IN   NS   eur5.akam.net.  
mit.edu.      172800  IN   NS   use5.akam.net.
```

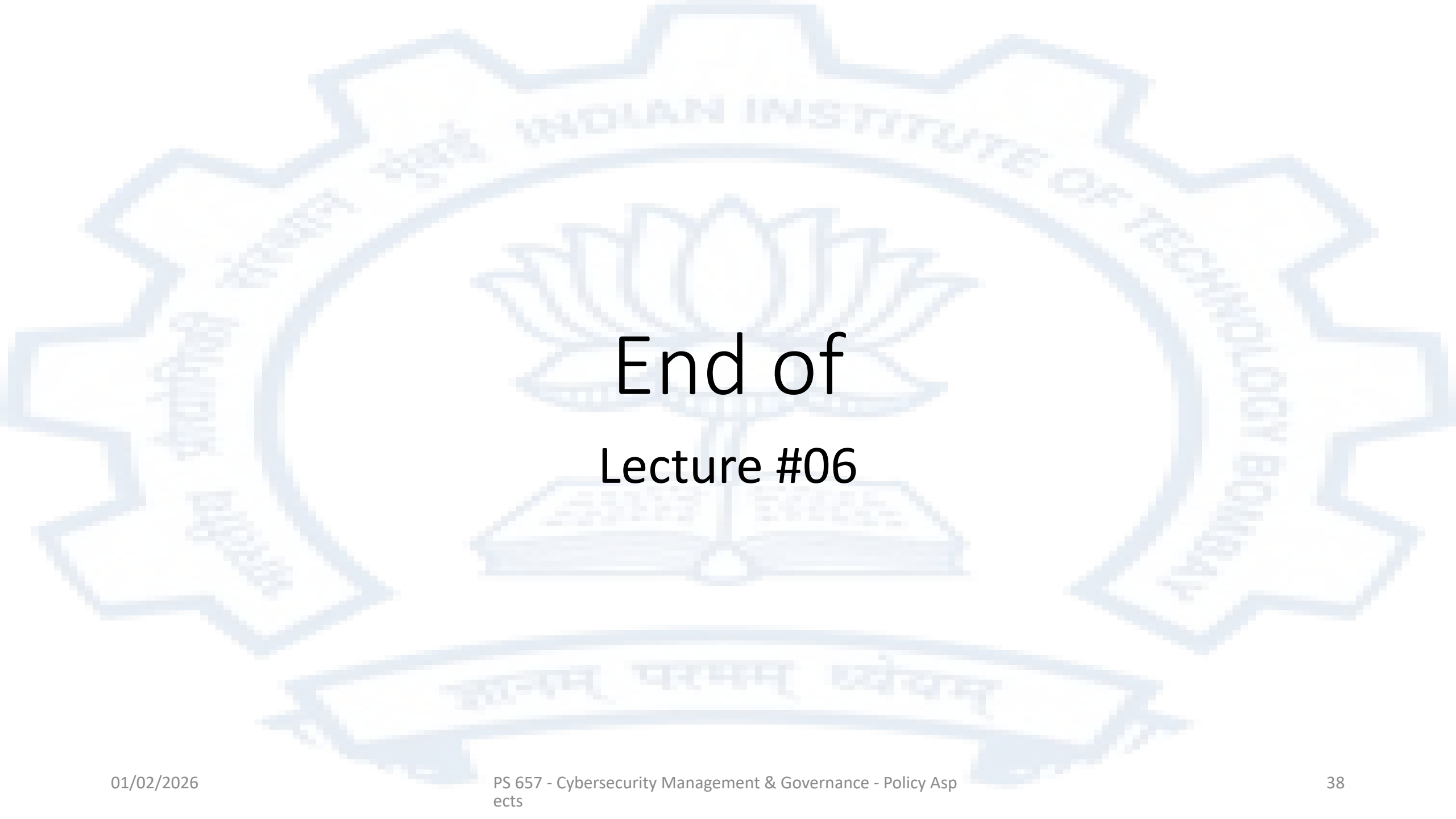
;; Received 409 bytes from 192.35.51.30#53(192.35.51.30) in 808 ms

```
mozg.mit.edu.  1800   IN   A    18.93.3.229
```

;; Received 46 bytes from 23.74.25.64#53(23.74.25.64) in 154 ms

To summarize

- Computer to computer communications required the consideration of a very different kind of communications network as opposed to traditional telecom networks that were mainly voice networks
- These newer techniques were based on switching packets (routing data) as opposed to creating a dedicated circuit between the communications endpoints
- TCP/IP rapidly became the most popular and led to the evolution of several interconnected networks that collectively are today known as the Internet
- The concept of an “address” is fundamental to the operation of such networks
- The enormous success of the Internet has led to an exhaustion of the older IPv4 addresses and the development of IPv6 was motivated by this. IPv6 seems to have enough addresses to last for a very long time
- In order to address usability and to provide flexibility the ability to use names instead of addresses is important and has led to the evolution of the Domain Name System (DNS)
- DNS resolution is now one of the most important and critical parts of the fundamental operation of the Internet.
- Governing the Internet is not as easy as the hierarchical devolution of control to countries as in traditional telecommunication systems

The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay (IIT Bombay) logo. The logo is a gear-like shape with a lotus flower in the center. The text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written along the top inner edge of the gear, and "सत्यमेव जयते" is written along the bottom inner edge.

End of Lecture #06